



RU-net: A Residual U-net for Automatic Interplanetary Coronal Mass Ejection Detection

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Abstract

Detection methods for interplanetary coronal mass ejections (ICMEs) from in situ spacecraft measurements are mostly manual, which are labor-intensive and time-consuming, being prone to the inconsistencies of identification criteria and the incompleteness of the existing catalogs. Therefore, the automatic detection of ICMEs has aroused the interest of the astrophysical community. Of these automatic methods, the convolutional neural network–based methods show the advantages of fast speed and high precision. To further improve the computing feasibility and detection performance, this paper proposes a method called residual U-net (RU-net), from the perspective of time-series segmentation. With the help of U-net architecture, we design an encoder–decoder network with skip connection to capture multiscale information, where the end-to-end architecture with an embedded residual element is formulated to accelerate the algorithmic convergence. For the in situ data from 1997 October 1 to 2016 January 1 collected by the Wind spacecraft, the results of our experiments demonstrate the competitive performance of the proposed RU-net in terms of accuracy and efficiency (178 of 230 ICMEs are detected in the test set, and the $F1$ score is 80.18%).

Unified Astronomy Thesaurus concepts: [Astronomy data analysis \(1858\)](#); [Solar coronal mass ejections \(310\)](#); [Computational methods \(1965\)](#)

1. Introduction

Coronal mass ejections (CMEs) are frequent and violent explosive phenomena in solar activity (O’Kane et al. 2021). During its eruption, the Sun ejects coronal mass, with high amounts of energy being released, which change the structure of the coronal layer in the eruption area significantly (Žic et al. 2015). If the CMEs propagate into interplanetary space during an explosion, the parts in interplanetary space are called interplanetary coronal mass ejections (ICMEs; Cane & Richardson 2003). Due to their characteristics of high speed, the strong southern component of the magnetic field, and long duration (Shen et al. 2017), ICMEs are considered as the major drivers of space weather effects and the main source of solar terrestrial space disturbance (Žic et al. 2015). In general, ICMEs may affect the normal state of the ionosphere around the Earth (Bothmer & Schwenn 1994; Rust 1994), trigger high-energy particle events (Berdichevsky et al. 2002; Laine & Aila 2016), and have a particular impact on space exploration, satellite communication, and power grids and power facilities within one to three days (Bilenko 2020).

With the launch of the spacecraft WIND and ACE, it is possible to permanently collect space data for monitoring ICMEs at the Lagrangian point L1. They allow the scientific community to study the physical characteristics, geomagnetic effects, and other corresponding events of ICMEs. Figure 1 shows the data observed by WIND from 1997 October 10 to

13, where the dashed vertical lines represent the start and end times of an ICME event. The first panel presents an enhanced and smoothly rotating B , which is mainly due to the existence of flux ropes inside the ICMEs with magnetic cloud structures. The second panel shows that the velocity initially increases, then decreases slowly when the shock related to the CME arrives. Meanwhile, the thermal velocity V_{th} (third panel), plasma parameters β (fourth panel), and density of protons N_p , $N_{p,nl}$, and $N_{\alpha,nl}$ (fifth panel) are also relatively small. The above signatures are the main criteria commonly used to identify ICMEs (Chi et al. 2016; Kilpua et al. 2017). However, not all ICMEs are preceded by a turbulent region sheath and/or a fast upstream shock as shown in Figure 1.

By visual inspection, Lepping et al. (2006) listed 82 magnetic clouds (MCs) between 1995 and 2003, and summarized various aspects of the results, such as a discussion of how an MC’s front boundary is specifically identified. Richardson & Cane (2010) listed 373 ICMEs between 1996 and 2009, and summarized their basic properties and geomagnetic effects. Jian et al. (2006) listed 230 ICMEs between 1995 and 2004, and examined the solar cycle variation of some properties of the ICMEs, such as occurrence rate and scale size. Nieves-Chinchilla et al. (2018) listed 337 ICMEs between 1995 and 2015, of which 298 showed organized magnetic field signatures. Chi et al. (2016) listed 465 ICMEs from 1995 to 2015, and showed that some properties of ICMEs, such as the number of ICMEs and the magnetic field, are well correlated with the solar cycle variation. On average, 80% of the ICMEs of a given list are presented in another list (Chi et al. 2016). However, there are ambiguities in this time-consuming task (Jian et al. 2006; Lepping et al. 2006; Zurbuchen & Richardson 2006; Richardson & Cane 2010; Chi et al. 2016; Kilpua et al. 2017; Nieves-Chinchilla et al. 2018). Some experts do not fully agree with the above ICME standards, which lead to

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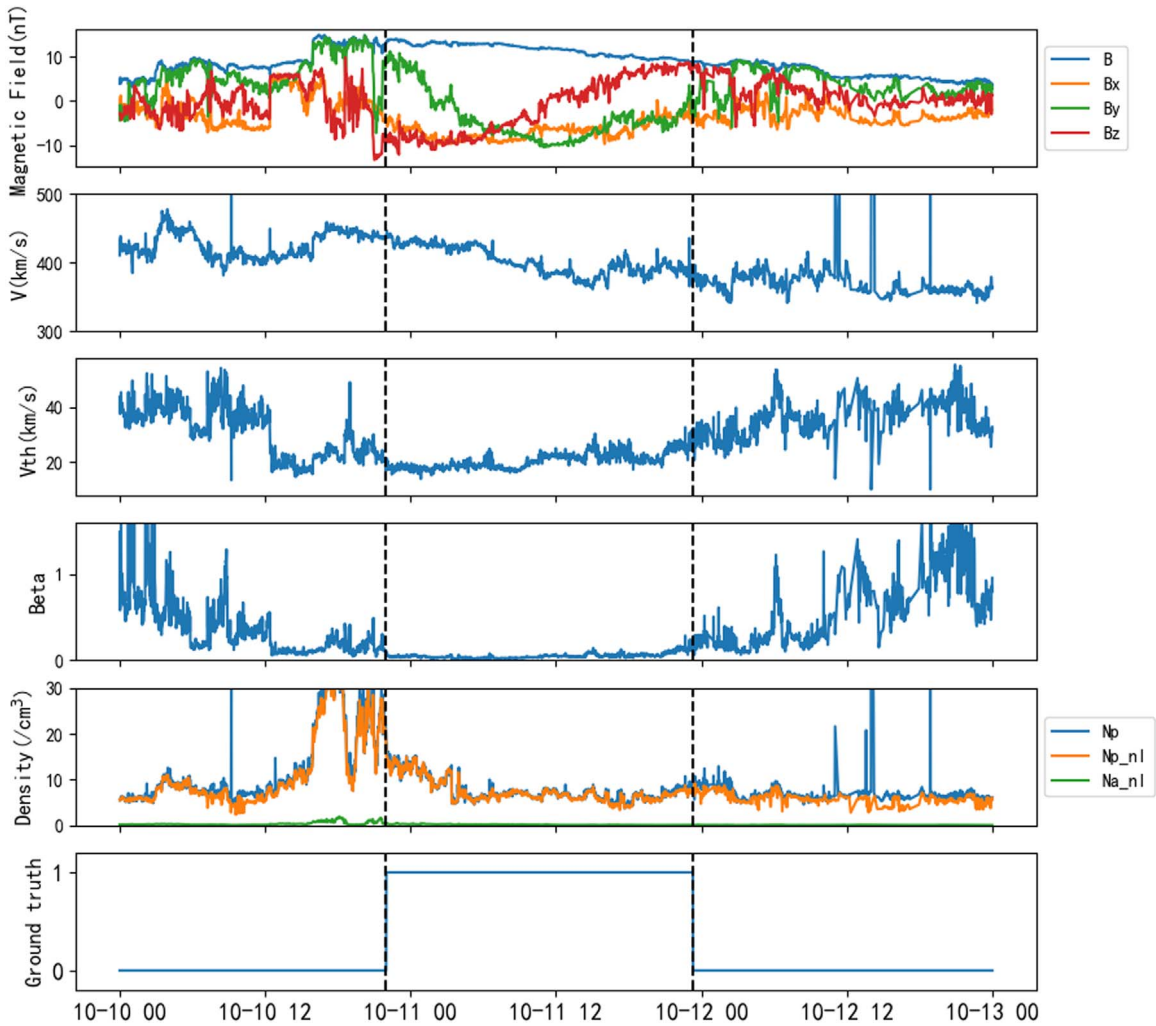


Figure 1. A demonstration of part of the data from 1997 October 10 to 13, including an ICME with start and end times indicated by the dashed vertical lines. From the top to the bottom, the first five panels represent the magnetic field B and its components B_x , B_y , and B_z ; the solar wind velocity V ; the thermal velocity V_{th} ; the plasma parameters β ; and the proton and α particle density obtained by moment analysis and nonlinear analysis, separately N_p , $N_{p,nl}$, and $N_{\alpha,nl}$. The last panel shows the real label.

the discrepancies between these ICME catalogs. For example, Nieves-Chinchilla et al. (2018) regarded the increase of plasma parameters as a characteristic of ICMEs, but Chi et al. (2016) did not. Despite all this, these ICME catalogs are still the basis for the statistical study of the physical properties of ICMEs, and they promote the development of an automatic detection algorithm.

To this end, the automatic identification method holds great promise for considerable gains in time and objectivity in relation to the elaboration of catalogs. Currently, many useful methods have been developed to address this problem. Lepping et al. (2005) proposed an automatic detection method of ICMEs based on existing empirical standards, which successfully detected some events but more false positive events. Ojeda-Gonzalez et al. (2017) identified candidate regions of ICMEs in a semiautomatic way by calculating spatio-temporal entropy, achieving a high precision of 70%. This achieved satisfactory results in a relatively small test set (9 of the 13 ICMEs were identified), which is considered to have great contingency. In addition, this method is based on a set of specific target events, which may fail on other data sets.

Over the past years, deep learning methods have achieved great advances in space weather forecasting, including the classification of solar flares (Nishizuka et al. 2018; Li et al. 2020) and the prediction of sunspot numbers (Pala & Atici 2019; Lee 2020). For ICME detection, Nguyen et al. (2019) provided a multiscale convolution neural network (CNN) method. The whole pipeline consists of three steps. First, the data are resampled to a 10 minute resolution, to eliminate most of the missing data while still maintaining accuracy in determining the start and end times of the labeled events. Then, 100 CNNs are trained on the resampled training data that are grouped into 100 sizes to predict the possibilities of the samples belonging to one ICME event on different scales. Finally, median filtration and a peak detection method are utilized to process the raw predictions of the 100 CNNs for converting the prediction possibilities into an ICME catalog.

Inspired by image segmentation, deep learning methods have also already been applied for the analysis of time series within various disciplines. For example, Lea et al. (2016) introduced a temporal convolutional network for action segmentation on video or other time-series data, which showed competitive performance on several data sets with lower time complexity.

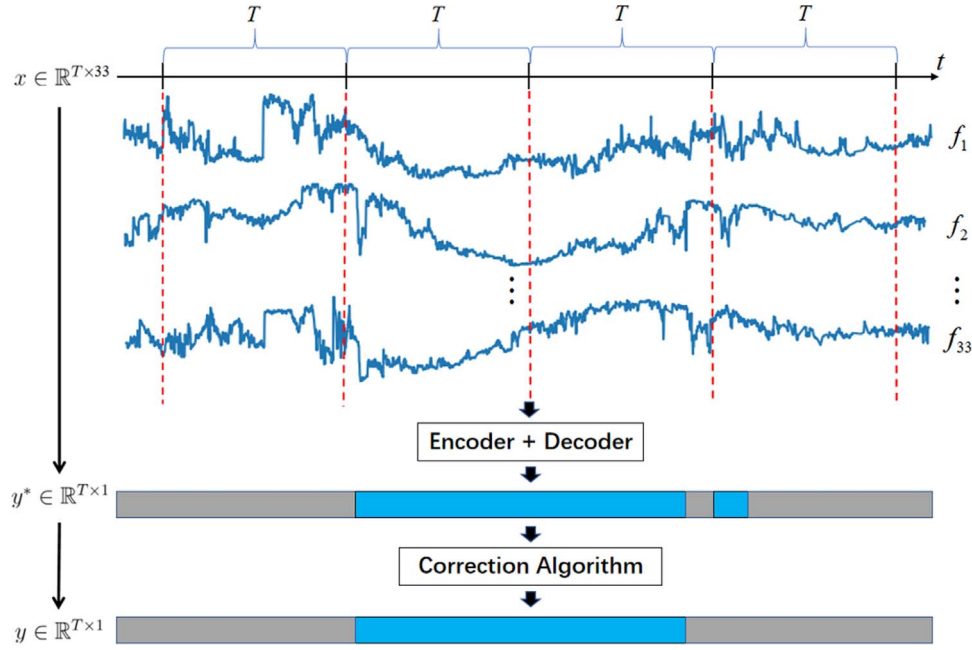


Figure 2. An illustration of the proposed pipeline. $\{f_i, i = 1, \dots, 33\}$ represents the feature vectors set, including 33 features, and x indicates the input. There are only 0 or 1 in the original output y^* and final output y (“0” and “1” represent non-ICMEs and ICMEs, respectively).

Doukhan & Carrive (2017) applied CNN to speech and music segmentation and classification, and this showed a significant improvement over the support vector machine and Gaussian Mixture Model. Perslev et al. (2019) proposed U-time, which is a more robust temporal fully convolutional network, based on the structure of U-net (Ronneberger et al. 2015), to achieve better results in the sleep phase segmentation mission. Jang et al. (2019) designed a new CNN with a Mel-scale kernel for music detection from broadcasting content. Gimeno et al. (2020) proposed a system for multiclass audio segmentation, based on bidirectional long short-term memory networks complemented by a resegmentation module. Dmitrieva et al. (2020) improved U-net to make it suitable for tasks in 1D, and then segmented the trajectory of protein to analyze its dynamics.

Though Nguyen et al. (2019) has shown excellent performance, there is still much room for improvement in terms of accuracy and efficiency. From the perspective of time-series segmentation, this paper proposes an encoder-decoder network with residual module (RU-net) to realize the end-to-end detection of ICMEs. The aim of time-series segmentation is to provide a compact representation by dividing the series into segments. Compared with the baseline (Nguyen et al. 2019), the two main advantages of this paper are listed as follows:

1. *Refined framework:* we go far beyond the traditional detection framework and propose to detect ICMEs from the perspective of time-series segmentation.
2. *High performance:* we significantly improve the performance and efficiency of ICME detection. The $F1$ score of RU-net can achieve an improvement of up to 5.68%, while the training time is reduced to 5.5 hr from 83.5 hr on the same data.

The rest of this paper is organized as follows. Section 2 introduces the data and formulates the whole pipeline. The results of the experiments and discussion are provided in

Sections 3 and 4, respectively. Finally, Section 5 draws the conclusions.

2. Data and Pipeline

In this section, we introduce the data and the two steps of the pipeline that enables automatic ICME detection from WIND in situ data measurements. An illustration of our pipeline is shown in Figure 2. The idea of our method is to predict a label (“0” or “1,” representing non-ICMEs and ICMEs, respectively) for each sample to distinguish whether it belongs to an ICME, and to regard successive samples with the label 1 as a predicted ICME.

2.1. Data

In this paper, we use the spatial environment data collected by the WIND spacecraft from 1997 October 1 to 2016 January 1, and an ICME catalog (including 657 ICMEs) that is derived from the integration of five existing ICME directories (Jian et al. 2006; Lepping et al. 2006; Richardson & Cane 2010; Chi et al. 2016; Nieves-Chinchilla et al. 2018). The data and the catalog can be found online.⁷ Based on its consistency with other confirmed directories, the ICME catalog is valid by default. The data consist of 33 features, including the bulk velocity V and its components V_x , V_y , and V_z ; the thermal velocity V_{th} ; the magnetic field B , its components B_x , B_y , and B_z , and their root mean squares σ_x , σ_y , and σ_z ; the density of protons and α particles obtained from both moment and nonlinear analysis N_p , $N_{p,nl}$, and $N_{\alpha,nl}$; 15 channels of proton flux (the values are between 0.3 keV and 10 keV); and three addition features (the plasma parameters $\beta = P_{th}/P_B$, where P_{th} and P_B indicate the thermal pressure and the magnetic pressure separately, the dynamic pressure $P_{dyn} = N_p V^2$, and the normalized magnetic fluctuation $\sigma_B = \sqrt{\sigma_{B_x}^2 + \sigma_{B_y}^2 + \sigma_{B_z}^2}/B$).

⁷ <https://github.com/gautiernguyen/Automatic-detection-of-ICMEs-at-1-AU-a-deep-learning-approach>

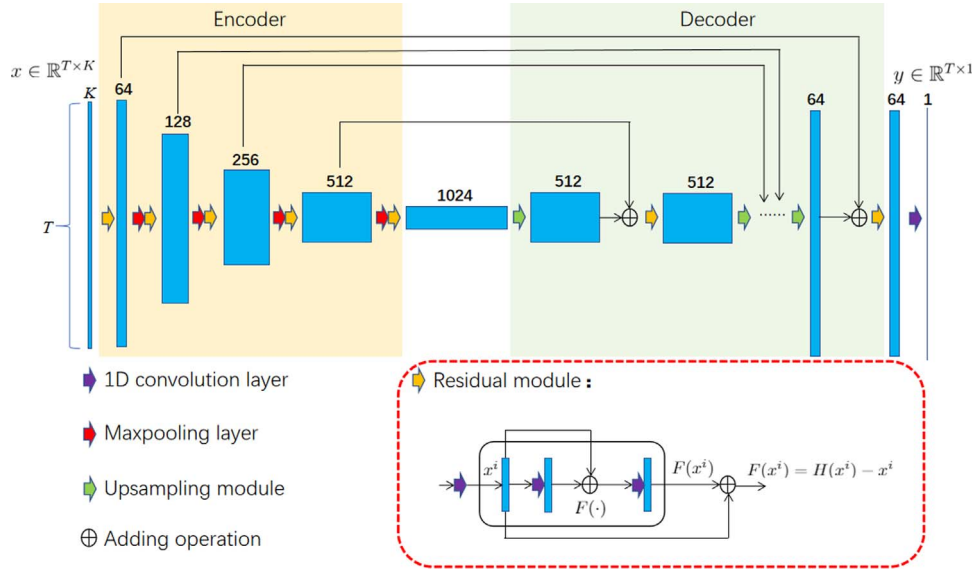


Figure 3. The overview of the proposed RU-net, where the purple arrow indicates the 1D convolution layer with kernel size 1×1 and sigmoid activation function, the red arrows denote the max pooling layer, the green arrows represent the upsampling module, and the yellow arrows indicate the residual module. The parts of the encoder and the decoder are presented in the light yellow and light green regions, respectively. The structure of the residual module is shown in the red dotted box, where x_i indicates the input of the residual block, $F(\cdot)$ and $F(x_i)$ represent the learned mapping and output of the convolution module separately, and $H(\cdot)$ indicates the output of the residual block.

In the following, each feature of the original data is scaled and normalized with an average of 0 and a standard deviation of 1 to avoid differences in orders of magnitude.

2.2. RU-net

Adopting the basic concept of U-net for 1D temporal segmentation, RU-net maps an input sequence to a segment as the output of a forward pass. Let $x \in \mathbb{R}^{T \times 33}$ be an input consisting of T consecutive samples with 33 channels. The RU-net model $\mathbb{R}^{T \times 33} \rightarrow \mathbb{R}^{T \times 1}$ gives each sample a prediction label. As shown in Figure 3, the architecture of RU-net is a fully convolutional encoder-decoder network, which is similar to the popular U-net architecture originally proposed for image segmentation, except that the convolutional blocks are replaced by residual blocks (He et al. 2015).

Encoder: the encoder f_{enc} (see the light yellow region in Figure 3) includes five residual blocks and four max pooling layers arranged in an alternative manner. It abstracts each normalized input x into a feature map, and these are retained for subsequent decoders to predict whether the samples contained belong to ICME events.

In the encoder f_{enc} , the number of filters in the five residual modules doubles after each max pooling operation, varying from 64 to 256. Max pooling (1) drastically reduces the computational and memory requirements; (2) enforces the learning of abstract features in the bottom layers; and (3) provides a large receptive field at the last convolution layer of the encoder. Besides, all convolutions in these blocks preserve the input dimensionality through zero padding. The aggressive max pooling reduces the input dimensionality by a factor of two. To ensure all four max pooling operations in the encoder, T is needed to be at least larger than 2^4 . If T is not large enough to make the long-range temporal relationship available, the performance of the model may be poor.

Decoder: the decoder f_{dec} (see the light green region in Figure 3) consists of four upsampling modules and four

residual blocks. It learns a mapping from the feature maps outputted by f_{enc} to the dense and pointwise segmentation.

In the encoder f_{dec} , the feature maps are first upsampled with a factor of two. Then, the feature maps together with the corresponding feature maps output by the encoder are input into the residual module. After four rounds, the number of channels in the feature maps is restored from 512 to 64. Finally, pointwise convolution (kernel size 1, filter size 1) with the sigmoid activation function will obtain the possibility that a sample belongs to an ICME event for each sample of the input x .

In summary, the whole network (encoder and decoder) maps the $T \times 33$ input to the $T \times 1$ confidence score, which is the final prediction result.

Residual block: as shown at the bottom of Figure 3, the residual block consists of three convolution layers and two shortcut connections. The reason why we use the residual structure here is that it greatly eases the training, on the premise of ensuring the performance.

For the i th residual module, the input is first input into a convolution layer with kernel size 1 and the output denoted as x^i . The sum of x^i and its output from the convolution layer with kernel size 3 is then input into the last convolution layer of the same kernel size. $F(\cdot)$ and $F(x^i)$ represent the learned mapping and output feature maps, respectively, and $H(x^i)$ indicates the output of the residual block.

The shortcut connection gives the derivative an identity term, which maintains the ability to backpropagate effectively and greatly promotes the optimization of the network. The main reason for the higher efficiency of residual block training is that the shortcut connection can make better use of information in different scales.

2.3. Correction Strategy

In the previous subsection, the RU-net indexes the intervals of ICME candidates from the ambient solar wind background, and Figure 4(b) illustrates a schematic diagram from 1997

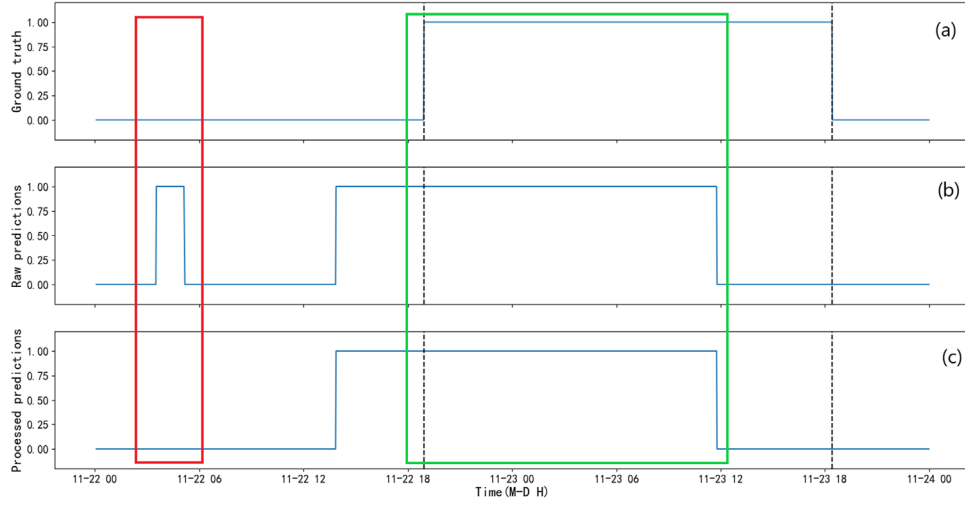


Figure 4. A schematic diagram of the correction algorithm. From the top to the bottom are the ground truth, the raw predictions, and the processed predictions from 1997 November 22 to 24. The start and end times of the ICMEs are indicated by the dashed vertical lines. The difference and the overlap among the ground truth, the raw, and the processed predictions are denoted by the red box and the green box, respectively.

November 22 to 24. Compared with the ground truth (in Figure 4(a)), the raw prediction contains two kinds of noise, such as close-by events and short-duration events.

Inspired by Nguyen et al. (2019), a simple correction strategy is adopted to improve the raw predictions. According to the prior statistics of the training data, we know that each ICME event lasts for at least 2 hr, and the time difference between adjacent events is at least 2 hr. Then, the following two corresponding correction strategies are adopted: (1) the predicted events with durations of less than 2 hr are removed; and (2) adjacent events with time differences of less than 2 hr are merged into one event. The final predicted ICME intervals are presented in Figure 4(c).

3. Experiments and Results

In this section, we first give a brief introduction to the setting of the experiments, and then show the results of the experiments to illustrate the effectiveness of the proposed pipeline.

3.1. Setting of Experiments

In order to illustrate the effectiveness of our model, this paper uses the same data partition as Nguyen et al. (2019); that is, the data from 1998–2009, 1997, and 2010–2016 are taken as the training set, the validation set, and the test set, respectively, so as to make the comparison fair.

The index intersection over ground truth (IoG) is chosen to judge whether each of the predicted ICMEs is true positive (TP):

$$\text{IoG}_{i,j} = \frac{\text{duration}(X_i \cap Y_j)}{\text{duration}(Y_j)}, \quad (1)$$

where X_i and Y_j represent the i th predicted ICMEs and the j th real ICMEs, respectively, and $\text{duration}(\cdot)$ indicates the duration of the event. A predicted ICME is defined as TP only if at least 10 consecutive samples are labeled as 1 in light of the concerns regarding the ambiguity of the border.

There may be two types of detection error, including false positive (FP) and false negative (FN). As shown in Table 1, an

Table 1
Confusion Matrix Classification

	Predicted ICMEs	Predicted non-ICMEs
ICMEs	TP	FN
Non-ICMEs	FP	TN

Table 2
The Selection of the Parameter *Ratio*

<i>Ratio</i>	<i>R</i>	<i>P</i>	<i>F1 Score</i>
0.5	0.6043	0.7989	0.6881
0.6	0.6217	0.7944	0.6976
0.7	0.6261	0.8090	0.7059
0.8	0.6087	0.8140	0.6951
0.9	0.5739	0.7253	0.6408
1.0	0.9913	0.1088	0.1960

Note. Bold values indicate optimal values.

FP is an ICME detected by our method but not on the ICME list, while an FN is an ICME on the ICME list that is not detected. On this basis, we employ the recall (R) and precision (P) to measure the learning pipeline. To be specific, R and P are defined as

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad P = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

Usually, there is a contradiction between R and P . Generally speaking, higher recall means lower precision, and vice versa. For comprehensive consideration, we also consider the $F1$ score that reveals the composite performance:

$$F1 = \frac{2 \times R \times P}{R + P}. \quad (2)$$

The ratio of positive samples to negative samples is 1:10. In order to alleviate the performance deterioration caused by serious data imbalance, we conducted experiments with U-net to determine the proportions of positive samples and negative samples in training. The results of the experiments are shown in Table 2, and the *Ratio* is chosen to be 0.7.

Table 3
The Detection Performance and Efficiency of Different Models

Model	10 Minute Sampling	Trainable Parameters	Training Time	Prediction Time	R	P	$F1$ score	IoG
Baseline (Nguyen et al. 2019)	✓	0.01M	83.5 hr	1.05 hr	0.7304	0.7602	0.7450	0.6207
Baseline	×	0.01M	>400 hr	5.27 hr	0.7783	0.7521	0.7650	0.5748
U-net	✓	9.77M	4.5 hr	0.07 hr	0.6261	0.8090	0.7059	0.6191
U-net	×	9.77M	4.5 hr	0.37 hr	0.8217	0.6974	0.7535	0.4767
RU-net	✓	13.61M	5.6 hr	0.06 hr	0.7609	0.7202	0.7400	0.6285
RU-net	×	13.61M	5.6 hr	0.35 hr	0.7739	0.8318	0.8018	0.4983
U-net2P	×	7.11M	4.2 hr	0.28 hr	0.8435	0.7293	0.7823	0.4859
U-net3P	×	25.14M	11.4 hr	0.5 hr	0.7391	0.8019	0.7692	0.3656
AttU-net	×	11.51M	5.5 hr	0.32 hr	0.8000	0.7244	0.7603	0.4367

Note. Bold values indicate optimal values.

An Adam optimizer with the learning rate 0.0001 is adopted to minimize the binary cross-entropy loss. The activation functions of the last layer and the other layers are the Sigmoid function and Relu function, respectively. All of the experiments are implemented on a personal computer with an i7-10750H @ 2.60 GHz Intel processor, and NVIDIA GeForce RTX 2070 GPU. The code is available online.⁸

3.2. Results of Experiments

In order to illustrate the effectiveness of our method, experiments are carried out on the following aspects: the missing value processing of data; the design idea for the training model; and the postprocessing of the prediction results.

3.2.1. Missing Value Processing

Note that there exist holes within the whole data set due to the instrumental constraints, and the great majority of these holes have durations between 2 and 10 minutes. Nguyen et al. (2019) proposed processing the original data by 10 minute sampling; that is, resampling the data to 10 minute resolution. Although this sampling strategy avoids the data analysis problems caused by the data defects, it will also lead to the loss of valuable information from the data, which is actually contrary to the original intention of improving the generalization of the model. Based on the above considerations, we adopt the strategy of filling the holes with 0 instead of the 10 minute sampling strategy.

We perform the ablation study of 10 minute sampling on the baseline (Nguyen et al. 2019), the U-net architecture (for time series), and the proposed RU-net architecture. The corresponding trainable parameters, training times, prediction times, and performances are reported in Table 3, where the performance of the baseline is determined by the decision threshold with the highest $F1$ score in Nguyen et al. (2019); ✓ means the 10 minute sampling, and × means not.

It is worth noting from Table 3 that the 10 minute sampling strategy greatly reduces the training time of the baseline from >400 to 83.5 hr, while sacrificing a little precision. However, instead of accelerating the training process for U-net and RU-net, the strategy degrades their performance, reducing the average $F1$ score by up to 5.47%. This is because the number of trainable parameters of the baseline is small, so the data processed by this strategy are enough to cover the simple architecture of the baseline, only two convolution blocks, one fully connected layer, and two maximum pool layers. In

contrast, the number of trainable parameters of 1D U-net and RU-net is several orders of magnitude larger than that of the baseline. These results suggest that this data preprocessing strategy is only suitable for algorithms with fewer parameters, and it is in line with the common sense understanding that sufficient data are helpful in improving the performance of deep learning models.

The experiments show that abandoning the 10 minute sampling strategy is conducive to improving the performance of RU-net, and that it achieves the highest $F1$ score of 80.18%.

3.2.2. Time-series Segmentation Framework

From Table 3, we can notice that the performance of U-net is slightly worse than the baseline, while RU-net is comparable with the baseline when the 10 minute sampling strategy is adopted to preprocess the data. Once this preprocessing strategy is discarded, the performance of U-net matches that of the baseline, and the performance of our RU-net significantly exceeds that of the baseline. Furthermore, to better show the superiority of the time-series segmentation framework, three U-net-based methods—U-net2P (Zhou et al. 2020), U-net3P (Huang et al. 2020), and U-net with an attention module (AttU-net; Oktay et al. 2018)—are used, which were originally developed for image segmentation but have been redesigned in this paper for 1D time-series segmentation.

Figure 5 presents the recall and precision of the different models, where the colored dots denote the time-series segmentation models and the black line represents the precision–recall curve of the baseline, published in Nguyen et al. (2019). We can notice that all of the colored dots are above the black curve line, which indicates the superiority of the time-series segmentation framework in ICME detection. RU-net and U-net3P are in the high precision area, and both have satisfactory recall rates. Meanwhile, U-net, U-net2P, and AttU-net are in the high recall area, and all of them have satisfactory precision rates. In addition, the trainable parameters, training times, and $F1$ scores are also reported in Table 3. We can see that the $F1$ scores of U-net, RU-net, U-net2P, U-net3P, and AttU-net all exceed that of the baseline. Besides, even the method with the largest training time of the five time-series segmentation architecture-based methods is 11.4 hr, less than one-seventh of the training time of the baseline.

The results of these experiments fully show that the method based on time-series segmentation has competitive advantages in terms of the performance and efficiency of ICME detection.

⁸ <https://github.com/JackieChan0808/time-series-segmentation>

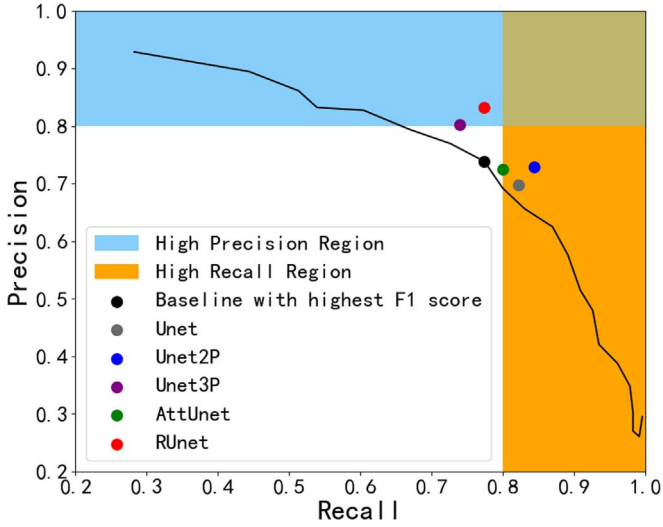


Figure 5. The detection results (P and R) of the different models. The blue zone and the yellow zone indicate the high precision region and the high recall region, respectively. The colored dots denote the time-series segmentation models, and the black line represents the precision–recall curve published in Nguyen et al. (2019).

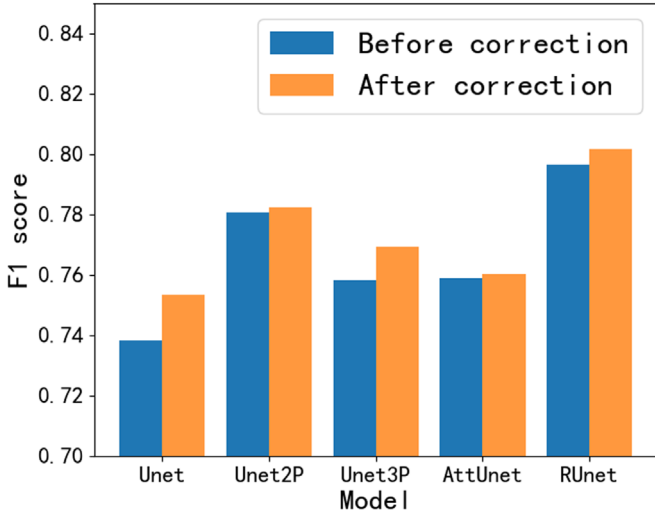


Figure 6. Ablation study using the correction algorithm.

Moreover, our method is superior to other U-net-based methods in terms of recall rate, precision, time complexity, and other factors.

3.2.3. Correction Strategies

Correction is a simple procedure to improve the raw predictions with the prior statistics from the training data. To show the effectiveness of the correction strategies, we conduct the ablation study on 1D U-net, RU-net, 1D U-net2P, 1D U-net3P, and AttU-net. The detection results ($F1$ scores) before and after the correction algorithm are visually presented in Figure 6.

It is worth noting from Figure 6 that there is a general improvement to the detection performance of each architecture. To be specific, the U-net architecture enjoys the largest improvement on the $F1$ score, up to 1.52%, while the AttU-net architecture has the smallest improvement, by 0.15%. In contrast, the $F1$ score of the proposed RU-net has increased by

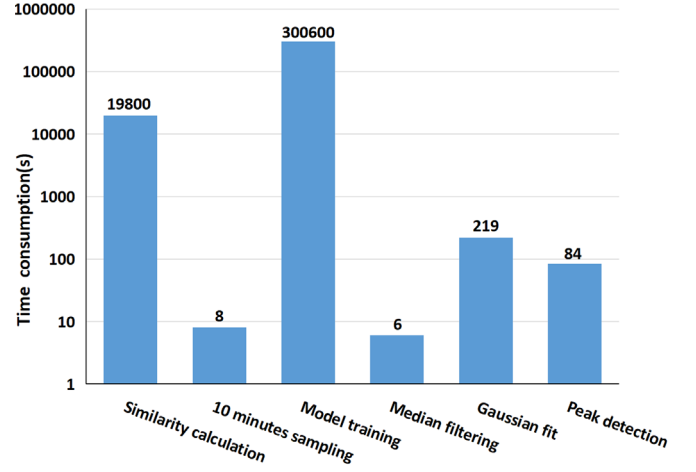


Figure 7. The time consumption of each module in the pipeline (Nguyen et al. 2019).

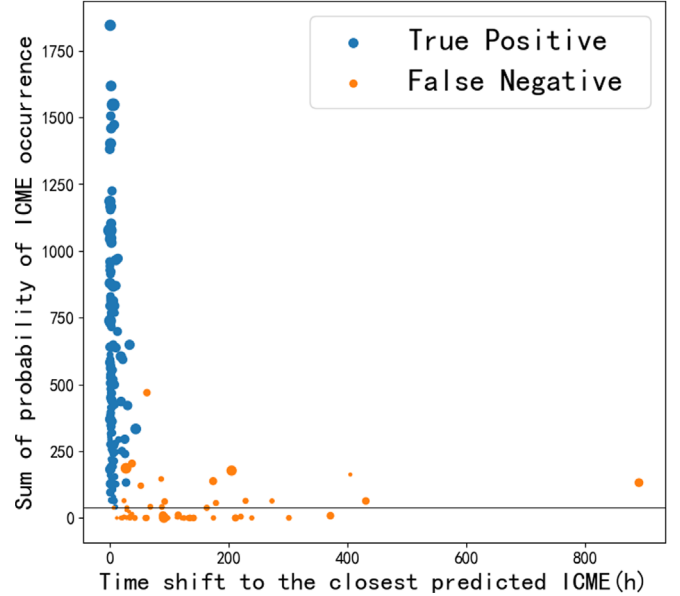


Figure 8. The sum of the probabilities of each of the ICMEs in our test period as a function of the temporal shift to the closest predicted ICMEs. The blue dots and yellow dots indicate TPs and FNs, respectively, while the size represents the duration. The black line indicates the minimum among the TPs.

0.54%. This supports the understanding that correction strategies can effectively improve the detection performance.

4. Discussion

In this section, we provide a detailed comparison with the baseline, and then give some statistical analysis of our results.

We show the time consumption of each module of the baseline in Figure 7, from which we can notice that the two steps of similarity calculation and model training are particularly time-consuming. In contrast, RU-net adopts an end-to-end structure, which omits the procedures other than model training. Furthermore, there is only one network that needs to be fitted in RU-net, taking 20,160 s, while there are 100 networks in the baseline, taking 300,600 s. Obviously, our algorithm significantly reduces the time complexity. In addition, the $F1$ score of RU-net is as high as 79.64%, even without the correction strategies, and the whole prediction

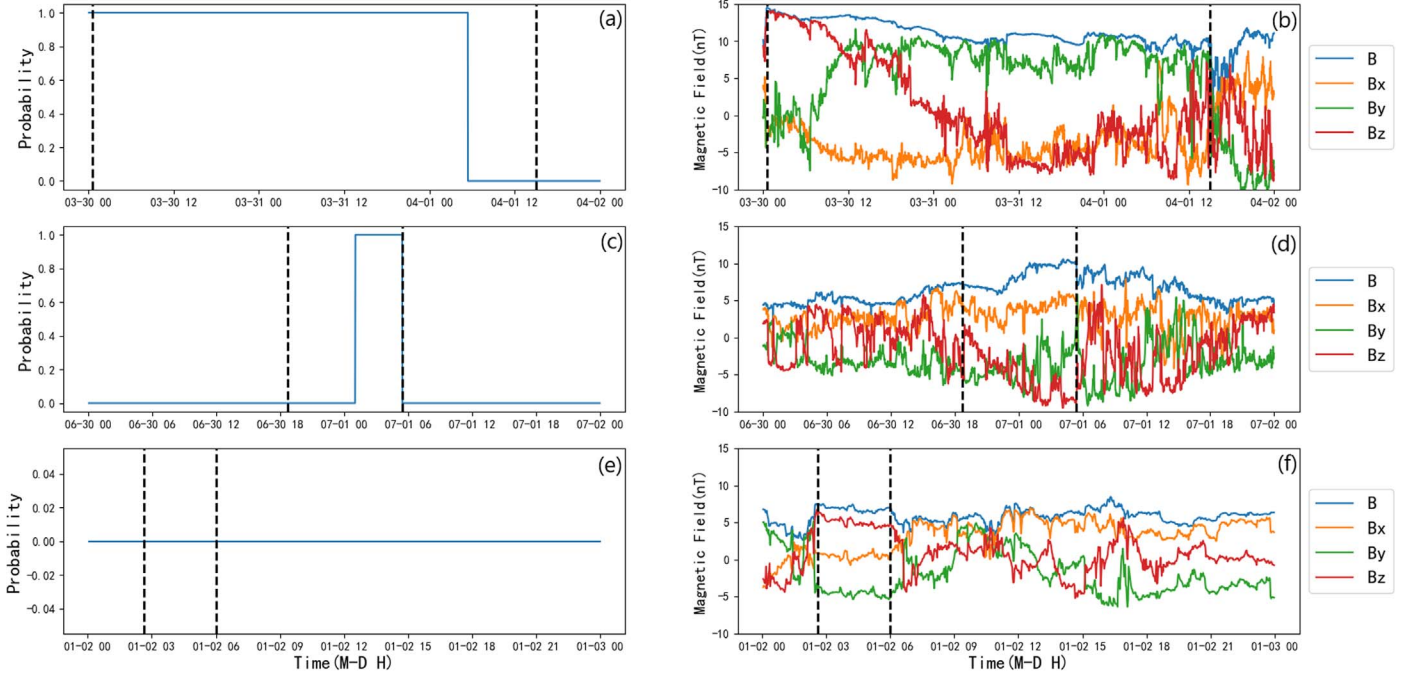


Figure 9. The predictions of a long TP (a), a short TP (c), and an FN (e), corresponding to the red dot in Figure 5, and their patterns on the magnetic field, as shown in (b), (d), and (f). The start and end times of these ICMEs are indicated by the dashed vertical lines.

procedure takes only about 2 s. But the two steps of Gaussian fitting and peak detection in Nguyen et al. (2019), as postprocessing procedures, need more than 5 minutes, and the sample sequence needs to be long enough, which inevitably limits its application in real-time scenarios. Considering that it takes 2~3 minutes for the spacecraft to collect a single data point, the proposed RU-net is no doubt more suitable for real-time monitoring of ICMEs.

The test set contains a total of 230 ICME events. We detected 214 events, of which 178 events are TPs. Figure 8 shows the sum of the probabilities that the data points contained in each ICME event are correctly predicted as a function of the time shift to the closest predicted ICME events. The size of each dot is proportional to the duration of the ICME event. The solid horizontal line indicates the minimum among the TPs. We can see that the sum of the probabilities varies over a large span, ranging from 41.0 to 1844.5. This may be because there is a positive correlation between the durations of TPs and the pattern of ICMEs in general (as shown in Figures 9(b) and (d)). With regard to the FNs, we can split them into two categories, according to the line. The FNs below the line are usually too weak in both their duration and pattern to be detected. The ones above the line are either too weak in duration or too weak in pattern, which are also hard to be detected (as shown in Figure 9(f)). In addition to the abovementioned variation curves of the magnetic fields of a long TP, a short TP, and an FN, we also draw the corresponding prediction curves (as shown in Figures 9(a), (c), and (e)). As a consequence, a great number of the FNs of our method are far from the predicted ICMEs, and thus correspond to very isolated events, which is consistent with the work done by Nguyen et al. (2019). Almost all of them have extremely short durations, and the rest all have extremely weak ICME patterns.

To explore the sources of the errors brought about by the proposed model, we perform statistical analysis on the FPs.

Because the ICME catalog we use is likely to be incomplete, there may be some FPs corresponding to ICMEs that have not been recorded. We visually present histogram statistics of the durations and sums of the probabilities of the samples during the period for TPs and FPs in Figure 10. In the left panel, all of the FPs have durations of less than 12 hr, roughly. The right panel shows that they should have sums of probabilities that are lower than those of the TPs. However, there may be some FPs that have similar patterns to ICMEs, considering the FPs with relatively long durations and high sums of probabilities, as well as the incompleteness of the ICME catalog. Studying these FPs will be beneficial to further supplement the existing ICMEs. An efficient way of distinguishing them would be given by the establishment of criteria according to these two parameters.

Besides, to ensure the physical reliability of our method, it is essential to study the consistency between the predicted ICMEs and the test ICMEs in terms of physical properties. Figure 11 presents their distributions of duration, magnet field, and thermal velocity, in which the left panels correspond to the predicted ICME list and the right panels correspond to the test ICME list. It can be seen that except for the slight differences in duration, the distributions of the other two indicators are almost the same, which fully illustrates that the predicted ICME list is consistent with the current existing catalogs.

To see if changing the training period has an influence on the performance of the pipeline, we train and evaluate our model with the data split in the same way as Nguyen et al. (2019). The *Recall*, *Precision*, and *F1* scores are shown in Table 4, from which we can notice that the performance of our algorithm deteriorates to a great extent, while the baseline shows better robustness. Changing the training period puts forward higher requirements for the robustness of the algorithm, which is one of the future directions of our study.

Finally, in order to test the robustness of the method and identify the features that play key roles in the automatic detection of ICMEs, we show the performance using the data

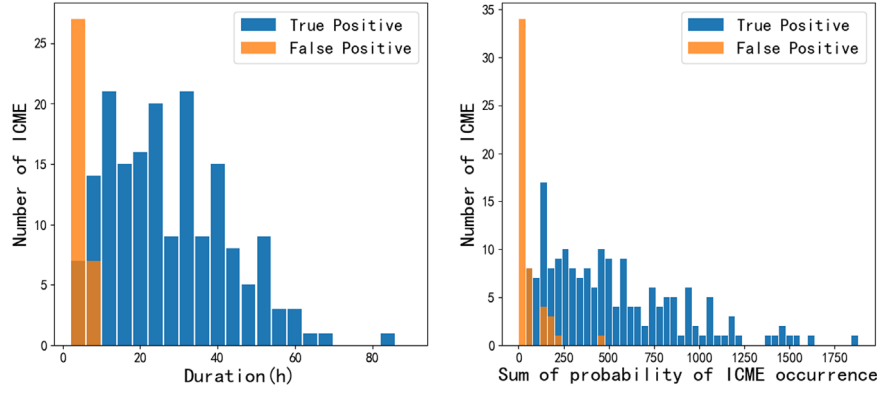


Figure 10. Histogram statistics of the duration (left) and sum of probability of the samples (right) during the period of TPs and FPs corresponding to RU-net (the red dot in Figure 5).

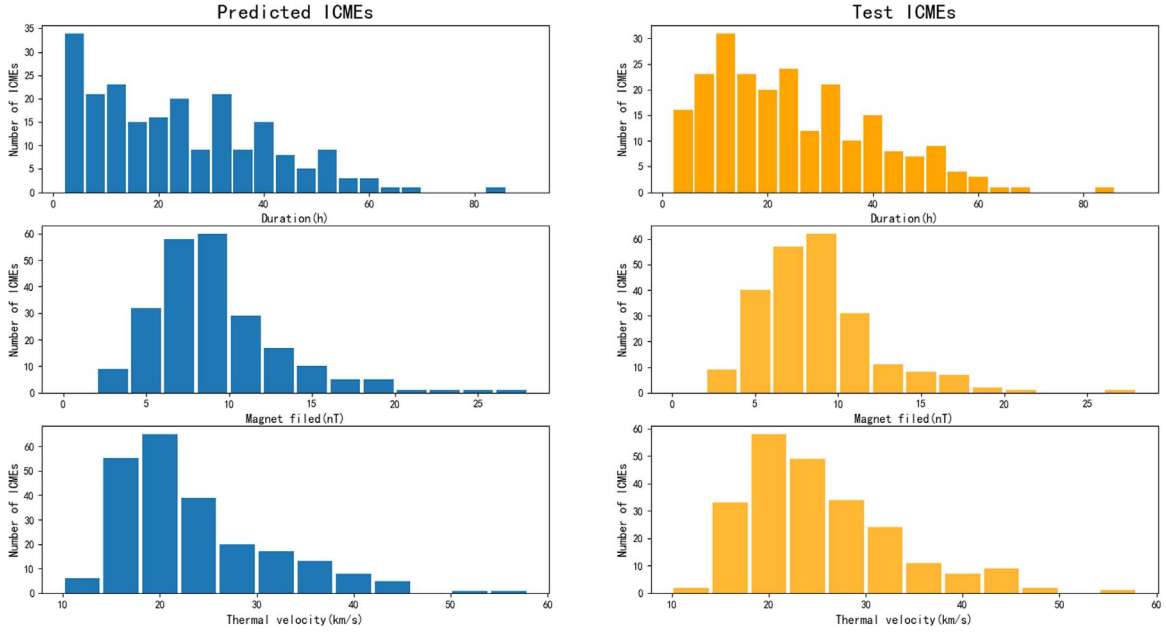


Figure 11. The distributions of duration, magnet field, and thermal velocity of the predicted ICMEs and the test ICMEs.

Table 4

The Performance of Our Method with Different Training Periods

Training Period	R	P	$F1$ Score
1998.1.1–2010.1.1	0.7739	0.8318	0.8018
2004.4.1–2015.12.31	0.8382	0.5575	0.6696
1997.10.1–2003.12.31	0.8374	0.4187	0.5583

Table 5

The Performance of Our Method with Different Data Set Features

Data Set Features	R	P	$F1$ Score
Complete data set	0.7739	0.8318	0.8018
B_x, B_y, B_z	0.8870	0.6163	0.7273
B_x, B_y, B_z, β , and proton fluxes	0.8304	0.6586	0.7346
Proton fluxes	0.8348	0.4683	0.6000

with absent features in the same configurations as Nguyen et al. (2019). As shown in Table 5 and Figure 12, the model using the data with whole features shows the best performance, which indicates that sufficient information is helpful to ICME detection. When only partial features are used to train the

network, the performance of the model deteriorates, especially when only information about proton fluxes is used, which indicates that features such as magnetic field and proton flux are of great significance for the automatic detection of ICME events.

5. Conclusions

Inspired by the pipeline in Nguyen et al. (2019) that uses CNNs to estimate a similarity for windows of data of various sizes and a postprocessing strategy based on peak detection, in this paper, we propose the pipeline RU-net, from the perspective of time-series segmentation, to automatically detect ICMEs from the WIND spacecraft's in situ measurements. Compared to the baseline, our pipeline adopts a more straightforward end-to-end framework, which in turn makes the data preprocessing and postprocessing steps more concise. The results of the experiments show that the $F1$ score of RU-net increases by 5.68%, while the training time is shortened from 83.5 to 5.5 hr. While improving the prediction accuracy, it greatly shortens the prediction time, which is of great significance to space weather forecasting.

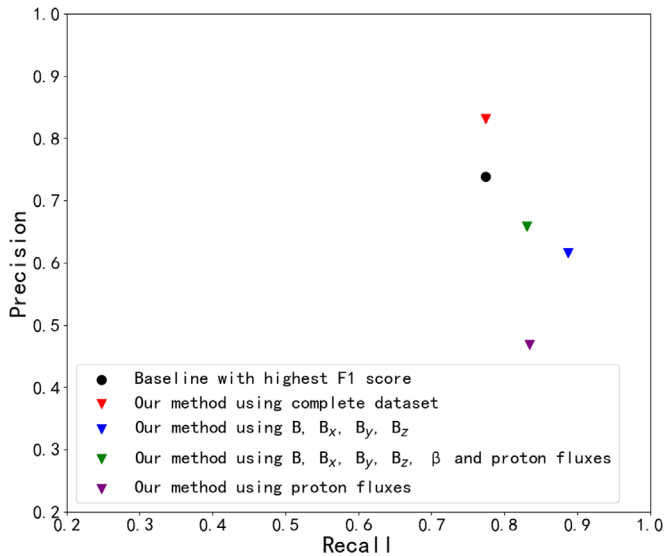


Figure 12. The performance of our pipeline using different data sets: the complete data set (the red nabla); only magnetic field information (the blue nabla); magnetic field, β , and proton flux information (the green nabla); and only proton flux information (the purple nabla). The black nabla represents the highest F1 score in Nguyen et al. (2019).

Along the lines of this paper, several open problems deserve further research. Mining more information from the data set is the first key step to designing a more effective method for ICME forecasting. However, due to the limitations of the data acquisition instruments or methods, CME data inevitably encounter the problem of data with missing features. Roughly discarding them will cause great waste, due to the high cost of the data acquisition. Besides, ICME events account for a small proportion of CME events. That is to say, we are facing the problem of data imbalance. Therefore, looking for data analysis methods that are consistent with the characteristics of the ICME data set for the better development of ICME forecasting pipelines is part of the future investigation.

The method also has plenty of other potential. The pipeline is designed without using any particular knowledge about ICME properties, so it can also be used to detect additional events such as stream interaction regions, corotating interaction regions, or ICME sheaths. Efforts can also be made to make the model lighter for application in online visualization tools or spacecraft. In addition, we can adapt this pipeline to the data of additional spacecraft in order either to increase the number of events in the frame of a statistical study or to track those events at different points of the interplanetary medium.

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